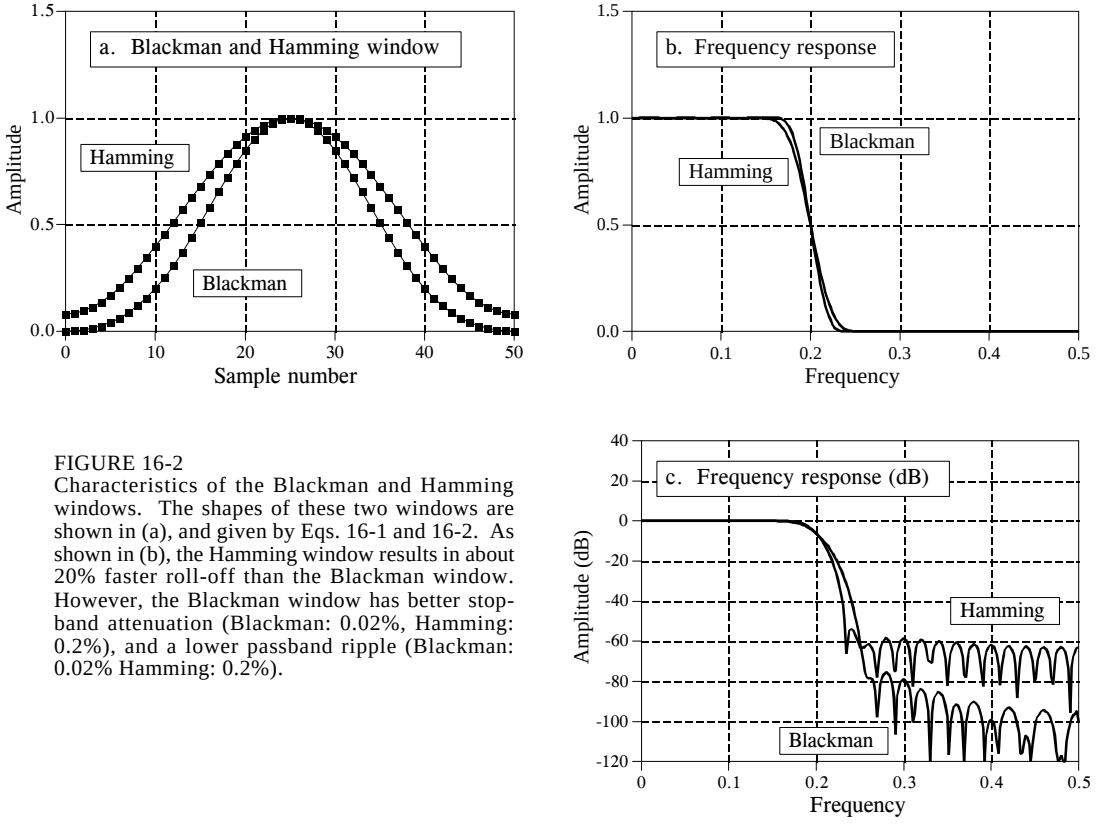

FIGURE 16-2
Characteristics of the Blackman and Hamming windows. The shapes of these two windows are shown in (a), and given by Eqs. 16-1 and 16-2. As shown in (b), the Hamming window results in about 20% faster roll-off than the Blackman window. However, the Blackman window has better stopband attenuation (Blackman: 0.02%, Hamming: 0.2%), and a lower passband ripple (Blackman: 0.02% Hamming: 0.2%).

(c) shows that the Blackman has a better *stopband attenuation*. To be exact, the stopband attenuation for the Blackman is -74dB (~0.02%), while the Hamming is only -53dB (~0.2%). Although it cannot be seen in these graphs, the Blackman has a *passband ripple* of only about 0.02%, while the Hamming is typically 0.2%. In general, the Blackman should be your first choice; a slow roll-off is easier to handle than poor stopband attenuation.

There are other windows you might hear about, although they fall short of the Blackman and Hamming. The **Bartlett window** is a triangle, using straight lines for the taper. The **Hanning window,** also called the **raised cosine window**, is given by: $w[i] = 0.5 - 0.5\cos(2\pi i/M)$. These two windows have about the same roll-off speed as the Hamming, but worse stopband attenuation (Bartlett: -25dB or 5.6%, Hanning -44dB or 0.63%). You might also hear of a **rectangular window**. This is the same as *no* window, just a truncation of the tails (such as in Fig. 16-1c). While the roll-off is ~2.5 times faster than the Blackman, the stopband attenuation is only -21dB (8.9%).

Designing the Filter

To design a windowed-sinc, two parameters must be selected: the cutoff frequency, f_C, and the length of the filter kernel, M. The cutoff frequency